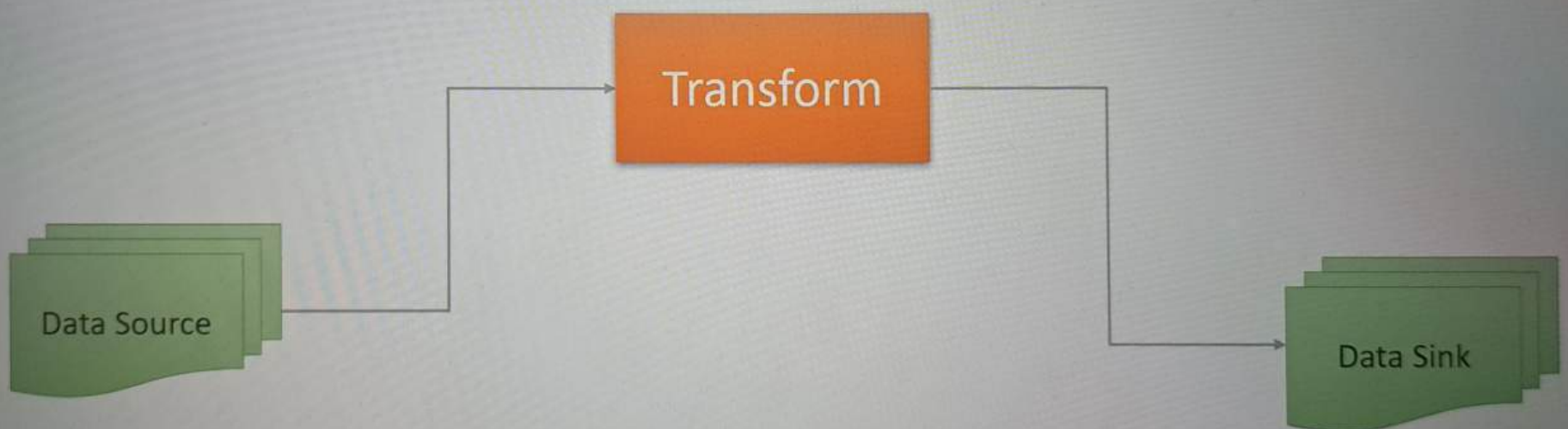
Introduction to Transformations



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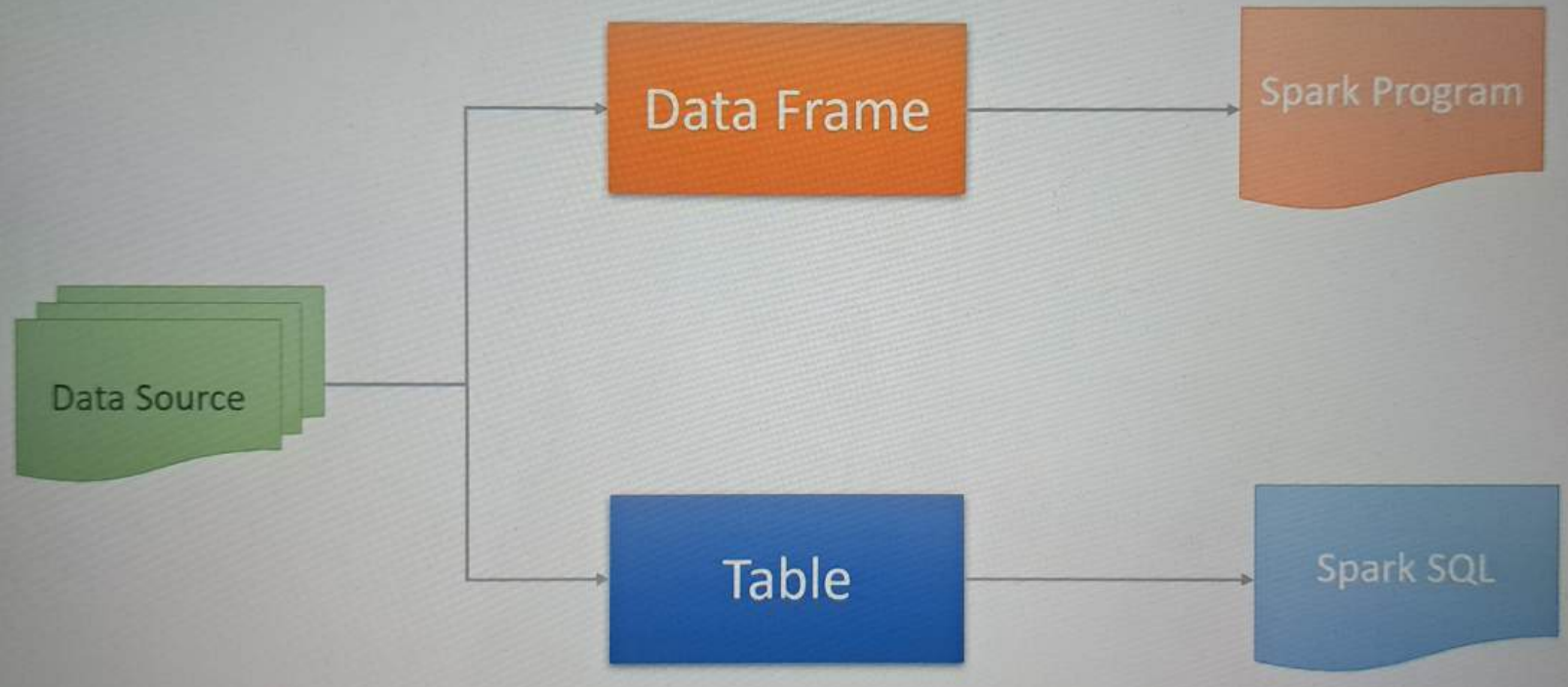
Introduction to Transformations



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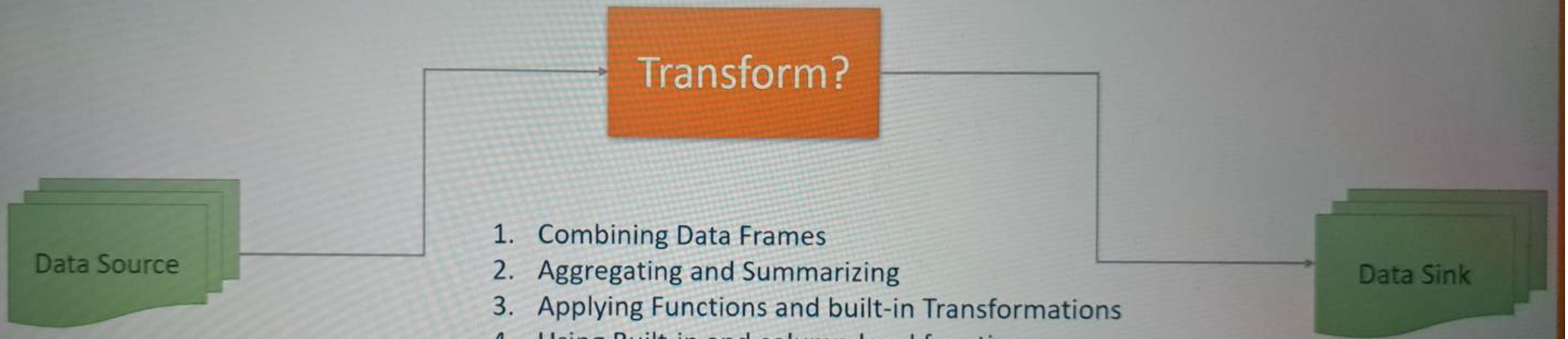
Introduction to Transformations



1. Combining Data Frames
2. Aggregating and Summarizing
3. Applying Functions and built-in Transformations
4. Using Built-in and column-level functions
5. Creating and using UDFs
6. Referencing Rows/Columns
7. Creating Column Expressions

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Working with Data Frame Row

Dataframe = Dataset[Row]

Timestamp	Age	Gender	Country	state
2014-08-27 11:29:31	37	Female	United States	IL
2014-08-27 11:29:37	44	M	United States	IN
2014-08-27 11:29:44	32	Male	Canada	
2014-08-27 11:29:46	31	Male	United Kingdom	
2014-08-27 11:30:22	31	Male	United States	TX

1. Manually creating Rows and Dataframe.
2. Collecting Dataframe rows to the driver.
3. Work with an individual row in Spark Transformations.

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Working with Data Frame Row

```
+---+----------+
| ID| EventDate|
+---+----------+
|123|04/05/2020|
|124|  4/5/2020|
|125| 04/5/2020|
|125| 4/05/2020|
+---+----------+
```

→

```
+---+----------+
| ID| EventDate|
+---+----------+
|123|2020-04-05|
|124|2020-04-05|
|125|2020-04-05|
|125|2020-04-05|
+---+----------+
```

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```python
from pyspark.sql import *
from pyspark.sql.functions import *
from pyspark.sql.types import *

def to_date_df(df, fmt, fld):
    return df.withColumn(fld, to_date(col(fld), fmt))
```

Command took 0.05 seconds -- by prashantkumarpandey@gmail.com at 6/16/2020, 7:46:01 PM on MyTestCluster

Cmd 2

```python
my_schema = StructType([
  StructField("ID", StringType()),
  StructField("EventDate", StringType())])

my_rows = [Row("123", "04/05/2020"), Row("124", "4/5/2020"), Row("125", "04/5/2020"), Row("126", "4/05/2020")]
my_rdd = spark.sparkContext.parallelize(my_rows, 2)
my_df = spark.createDataFrame(my_rdd, my_schema)

new_df = to_date_df(my_df,  "M/d/y", "EventDate")
```

Shift+Enter to run shortcuts

MyPythonNotebook - Databrick

community.cloud.databricks.com/?o=5086207005927225#notebook/2858727166054233/command/2858727166054236

MyPythonNotebook (Python)

MyTestCluster | File | Edit | View: Code | Permissions | Run All | Clear | Publish | Comments | Runs | Revision history

```
my_df.printSchema()
my_df.show()
new_df = to_date_df(my_df,  "M/d/y", "EventDate")
new_df.printSchema()
new_df.show()
```

(4) Spark Jobs

new_df: pyspark.sql.dataframe.DataFrame = [ID: string, EventDate: date]

```
root
 |-- ID: string (nullable = true)
 |-- EventDate: string (nullable = true)

+---+----------+
| ID| EventDate|
+---+----------+
|123|04/05/2020|
|124|  4/5/2020|
|125| 04/5/2020|
|126| 4/05/2020|
+---+----------+

root
 |-- ID: string (nullable = true)
 |-- EventDate: date (nullable = true)

+---+----------+
| ID| EventDate|
+---+----------+
|123|2020-04-05|
|124|2020-04-05|
|125|2020-04-05|
|126|2020-04-05|
+---+----------+
```

File Edit View Navigate Code Refactor Run Tools VCS Window Help 07-RowDemo - RowDemo_Test.py - PyCharm

07-RowDemo > RowDemo_Test.py

Project

07-RowDemo C:\Users\prashant\Documents
- app-logs
- lib
- log4j.properties
- RowDemo.py
- RowDemo_Test.py

External Libraries

Scratches and Consoles

RowDemo.py | RowDemo_Test.py

```python

from RowDemo import to_date_df


class RowDemoTestCase(TestCase):
    @classmethod
    def setUpClass(cls) -> None:
        cls.spark = SparkSession.builder \
            .master("local[3]") \
            .appName("RowDemoTest") \
            .getOrCreate()

        my_schema = StructType([
            StructField("ID", StringType()),
            StructField("EventDate", StringType())])

        my_rows = [Row("123", "04/05/2020"), Row("124", "4/5/2020"), Row("125", "04/5/2020"), Row("126", "4/05/2020")]
        my_rdd = cls.spark.sparkContext.parallelize(my_rows, 2)
        cls.my_df = cls.spark.createDataFrame(my_rdd, my_schema)

    def test_data_type(self):
        rows = to_date_df(self.my_df, "M/d/y", "EventDate").collect()
        for row in rows:
            self.assertIsInstance(row["EventDate"], date)

    def test_date_value(self):
        rows = to_date_df(self.my_df, "M/d/y", "EventDate").collect()
        for row in rows:
            self.assertEqual(row["EventDate"], date(2020, 4, 5))

```

9: Git | 6: TODO | 4: Run | Terminal | Python Console | Event Log

Packages installed successfully: Installed packages: 'pyspark==2.4.5' (16 minutes ago)

35:1 CRLF UTF-8 4 spaces Python 3.7 (pyspark45) master

Working with Data Frame Row

```
agg(*exprs)
cov(col1, col2)
crosstab(col1, col2)
cube(*cols)
filter(condition)
groupBy(*cols)
join(other, on=None, how=None)
orderBy(*cols, **kwargs)
replace(to_replace, value, subset)
rollup(*cols)
```

```
select(*cols)
sort(*cols, **kwargs)
where(condition)
withColumn(colName, col)
avg(*cols)
max(*cols)
mean(*cols)
min(*cols)
sum(*cols)
```

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File Edit View Navigate Code Refactor Run Tools VCS Window Help 09-LogFileDemo - apache_logs.txt - PyCharm

09-LogFileDemo > data > apache_logs.txt

Project

- 09-LogFileDemo C:\Users\prashant\Documents\GitHub\
  - app-logs
  - data
    - apache_logs.txt
  - lib
  - log4j.properties
  - LogFileDemo.py
- External Libraries
- Scratches and Consoles

apache_logs.txt

```
83.149.9.216 - - [17/May/2015:10:05:03 +0000] "GET /presentations/logstash-monitorama-2013/images/kibana-search.png HTTP/1.1" 200 203023
83.149.9.216 - - [17/May/2015:10:05:43 +0000] "GET /presentations/logstash-monitorama-2013/images/kibana-dashboard3.png HTTP/1.1" 200 171
83.149.9.216 - - [17/May/2015:10:05:47 +0000] "GET /presentations/logstash-monitorama-2013/plugin/highlight/highlight.js HTTP/1.1" 200 26
83.149.9.216 - - [17/May/2015:10:05:12 +0000] "GET /presentations/logstash-monitorama-2013/plugin/zoom-js/zoom.js HTTP/1.1" 200 7697 "htt
83.149.9.216 - - [17/May/2015:10:05:07 +0000] "GET /presentations/logstash-monitorama-2013/plugin/notes/notes.js HTTP/1.1" 200 2892 "http
83.149.9.216 - - [17/May/2015:10:05:34 +0000] "GET /presentations/logstash-monitorama-2013/images/sad-medic.png HTTP/1.1" 200 430406 "htt
83.149.9.216 - - [17/May/2015:10:05:57 +0000] "GET /presentations/logstash-monitorama-2013/css/fonts/Roboto-Bold.ttf HTTP/1.1" 200 38720
83.149.9.216 - - [17/May/2015:10:05:50 +0000] "GET /presentations/logstash-monitorama-2013/css/fonts/Roboto-Regular.ttf HTTP/1.1" 200 418
83.149.9.216 - - [17/May/2015:10:05:24 +0000] "GET /presentations/logstash-monitorama-2013/images/frontend-response-codes.png HTTP/1.1" 2
83.149.9.216 - - [17/May/2015:10:05:50 +0000] "GET /presentations/logstash-monitorama-2013/images/kibana-dashboard.png HTTP/1.1" 200 3216
83.149.9.216 - - [17/May/2015:10:05:46 +0000] "GET /presentations/logstash-monitorama-2013/images/Dreamhost_logo.svg HTTP/1.1" 200 2126 "
83.149.9.216 - - [17/May/2015:10:05:11 +0000] "GET /presentations/logstash-monitorama-2013/images/kibana-dashboard2.png HTTP/1.1" 200 394
83.149.9.216 - - [17/May/2015:10:05:19 +0000] "GET /presentations/logstash-monitorama-2013/images/apache-icon.gif HTTP/1.1" 200 8095 "htt
83.149.9.216 - - [17/May/2015:10:05:33 +0000] "GET /presentations/logstash-monitorama-2013/images/nagios-sms5.png HTTP/1.1" 200 78075 "ht
83.149.9.216 - - [17/May/2015:10:05:00 +0000] "GET /presentations/logstash-monitorama-2013/images/redis.png HTTP/1.1" 200 25230 "http://s
83.149.9.216 - - [17/May/2015:10:05:25 +0000] "GET /presentations/logstash-monitorama-2013/images/elasticsearch.png HTTP/1.1" 200 8026 "h
83.149.9.216 - - [17/May/2015:10:05:59 +0000] "GET /presentations/logstash-monitorama-2013/images/logstashbook.png HTTP/1.1" 200 54662 "h
83.149.9.216 - - [17/May/2015:10:05:30 +0000] "GET /presentations/logstash-monitorama-2013/images/github-contributions.png HTTP/1.1" 200
83.149.9.216 - - [17/May/2015:10:05:53 +0000] "GET /presentations/logstash-monitorama-2013/css/print/paper.css HTTP/1.1" 200 4254 "http:/
83.149.9.216 - - [17/May/2015:10:05:24 +0000] "GET /presentations/logstash-monitorama-2013/images/1983_delorean_dmc-12-pic-38289.jpeg HTT
83.149.9.216 - - [17/May/2015:10:05:54 +0000] "GET /presentations/logstash-monitorama-2013/images/simple-inputs-filters-outputs.jpg HTTP/
83.149.9.216 - - [17/May/2015:10:05:33 +0000] "GET /presentations/logstash-monitorama-2013/images/tiered-outputs-to-inputs.jpg HTTP/1.1"
83.149.9.216 - - [17/May/2015:10:05:56 +0000] "GET /favicon.ico HTTP/1.1" 200 3638 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_1) Ap
24.236.252.67 - - [17/May/2015:10:05:40 +0000] "GET /favicon.ico HTTP/1.1" 200 3638 "-" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:26.0)
93.114.45.13 - - [17/May/2015:10:05:14 +0000] "GET /articles/dynamic-dns-with-dhcp/ HTTP/1.1" 200 18848 "http://www.google.ro/url?sa=t&rc
93.114.45.13 - - [17/May/2015:10:05:04 +0000] "GET /reset.css HTTP/1.1" 200 1015 "http://www.semicomplete.com/articles/dynamic-dns-with-d
93.114.45.13 - - [17/May/2015:10:05:45 +0000] "GET /style2.css HTTP/1.1" 200 4877 "http://www.semicomplete.com/articles/dynamic-dns-with-
93.114.45.13 - - [17/May/2015:10:05:14 +0000] "GET /favicon.ico HTTP/1.1" 200 3638 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:25.0) Gecko/20
93.114.45.13 - - [17/May/2015:10:05:17 +0000] "GET /images/jordan-80.png HTTP/1.1" 200 6146 "http://www.semicomplete.com/articles/dynamic
93.114.45.13 - - [17/May/2015:10:05:21 +0000] "GET /images/web/2009/banner.png HTTP/1.1" 200 52315 "http://www.semicomplete.com/style2.cs
66.249.73.135 - - [17/May/2015:10:05:40 +0000] "GET /blog/tags/ipv6 HTTP/1.1" 200 12251 "-" "Mozilla/5.0 (iPhone; CPU iPhone OS 6_0 like
50.16.19.13 - - [17/May/2015:10:05:10 +0000] "GET /blog/tags/puppet?flav=rss20 HTTP/1.1" 200 14872 "http://www.semicomplete.com/blog/tags
66.249.73.185 - - [17/May/2015:10:05:37 +0000] "GET / HTTP/1.1" 200 37932 "-" "Mozilla/5.0 (compatible; Googlebot/2.1; +http://www.google
```

1. IP
2. client.
3. user
4. datetime
5. cmd
6. request
7. protocol
8. status
9. bytes
10. referrer
11. userAgent

9: Git 6: TODO Terminal Python Console

Event Log

1:48 LF UTF-8 4 spaces Python 3.7 (pyspark45) master

File Edit View Navigate Code Refactor Run Tools VCS Window Help 09-LogFileDemo - LogFileDemo.py - PyCharm

09-LogFileDemo | LogFileDemo.py

Project

- 09-LogFileDemo C:\Users\prashant\Documents\GitHub\
  - app-logs
  - data
    - apache_logs.txt
  - lib
  - log4j.properties
  - LogFileDemo.py
- External Libraries
- Scratches and Consoles

LogFileDemo.py

```python
from pyspark.sql import *
from pyspark.sql.functions import *

if __name__ == "__main__":
    spark = SparkSession \
        .builder \
        .master("local[3]") \
        .appName("LogFileDemo") \
        .getOrCreate()

    file_df = spark.read.text("data/apache_logs.txt")
    file_df.printSchema()
    log_reg = r'^(\S+) (\S+) (\S+) \[([\w:/]+\s[+\-]\d{4})\] "(\S+) (\S+) (\S+)" (\d{3}) (\S+) "(\S+)" "([^"]*)'

    logs_df = file_df.select(regexp_extract('value', log_reg, 1).alias("ip"),
                             regexp_extract('value', log_reg, 4).alias('date'),
                             regexp_extract('value', log_reg, 6).alias('request'),
                             regexp_extract('value', log_reg, 10).alias('referrer'))
    logs_df.printSchema()
```

if __name__ == "__main__"

Run: LogFileDemo

```
 |-- value: string (nullable = true)

root
 |-- ip: string (nullable = true)
 |-- date: string (nullable = true)
 |-- request: string (nullable = true)
 |-- referrer: string (nullable = true)

Process finished with exit code 0
```

9: Git | 6: TODO | 4: Run | Terminal | Python Console | Event Log

PEP 8: W292 no newline at end of file

19:26 CRLF UTF-8 4 spaces Python 3.7 (pyspark45) master

Data Frame Columns

Column

Timestamp	Age	Gender	Country	state
2014-08-27 11:29:31	37	Female	United States	IL
2014-08-27 11:29:37	44	M	United States	IN
2014-08-27 11:29:44	32	Male	Canada	
2014-08-27 11:29:46	31	Male	United Kingdom	
2014-08-27 11:30:22	31	Male	United States	TX

1. What is a Column and How to reference it?
2. How to create column expressions?

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ColumnsNotebook - Databricks

community.cloud.databricks.com/?o=5086207005927225#notebook/3635560979762597

ColumnsNotebook (Python)

MyTestCluster | File | Edit | View: Code | Permissions | Run All | Clear | Publish | Comments | Runs | Revision history

```
.option("inferSchema","true") \
.option("samplingRatio", "0.0001") \
.load("/databricks-datasets/airlines/part-00000")
```

(2) Spark Jobs

airlinesDF: pyspark.sql.dataframe.DataFrame = [Year: integer, Month: integer ... 29 more fields]

Command took 15.37 seconds -- by prashantkumarpandey@gmail.com at 6/21/2020, 12:58:00 PM on MyTestCluster

Cmd 5

Shift+Enter to run shortcuts

1. Column String
2. Column Object

Cmd 5

```
airlinesDF.select("Origin", "Dest", "Distance" ).show(10)
```

▸ (1) Spark Jobs

```
+------+----+--------+
|Origin|Dest|Distance|
+------+----+--------+
|   SAN| SFO|     447|
|   SAN| SFO|     447|
|   SAN| SFO|     447|
|   SAN| SFO|     447|
|   SAN| SFO|     447|
|   SAN| SFO|     447|
|   SAN| SFO|     447|
|   SAN| SFO|     447|
|   SAN| SFO|     447|
|   SAN| SFO|     447|
+------+----+--------+
only showing top 10 rows
```

Command took 0.97 seconds -- by prashantkumarpandey@gmail.com at 6/21/2020, 12:59:12 PM on MyTestCluster

Cmd 6

```
from pyspark.sql.functions import *
airlinesDF.select(column("Origin"), col("Dest"), airlinesDF.Distance ).show(10)
```

▸ (1) Spark Jobs

```
+------+----+--------+
|Origin|Dest|Distance|
+------+----+--------+
|   SAN| SFO|     447|
|   SAN| SFO|     447|
|   SAN| SFO|     447|
|   SAN| SFO|     447|
|   SAN| SFO|     447|
|   SAN| SFO|     447|
|   SAN| SFO|     447|
|   SAN| SFO|     447|
|   SAN| SFO|     447|
|   SAN| SFO|     447|
+------+----+--------+
only showing top 10 rows
```

How to create column expressions?

udemy

ColumnsNotebook - Databricks

community.cloud.databricks.com/?o=5086207005927225#notebook/3635560979762597

ColumnsNotebook (Python)

MyTestCluster | File | Edit | View: Code | Permissions | Run All | Clear | Publish | Comments | Runs | Revision history

Command took 15.37 seconds -- by prashantkumarpandey@gmail.com at 6/21/2020, 12:58:00 PM on MyTestCluster

Cmd 5

```
airlinesDF.select("Origin", "Dest", "Distance" ).show(10) ...
```

Show cell

Cmd 6

```
from pyspark.sql.functions import * airlinesDF.select(column("Origin"), col("Des ...
```

Show cell

Cmd 7

Shift+Enter to run shortcuts

1. String Expressions or SQL Expressions
2. Column Object Expressions

Cmd 7

```
airlinesDF.select("Origin", "Dest", "Distance", "Year","Month","DayofMonth").sho ...
```

Show cell

Cmd 8

```
airlinesDF.select("Origin", "Dest", "Distance", expr("to_date(concat(Year,Month,DayofMonth),'yyyyMMdd') as FlightDate")).show(10)
```

▸ (1) Spark Jobs

```
+------+----+--------+----------+
|Origin|Dest|Distance|FlightDate|
+------+----+--------+----------+
|   SAN| SFO|     447|1987-10-14|
|   SAN| SFO|     447|1987-10-15|
|   SAN| SFO|     447|1987-10-17|
|   SAN| SFO|     447|1987-10-18|
|   SAN| SFO|     447|1987-10-19|
|   SAN| SFO|     447|1987-10-21|
|   SAN| SFO|     447|1987-10-22|
|   SAN| SFO|     447|1987-10-23|
|   SAN| SFO|     447|1987-10-24|
|   SAN| SFO|     447|1987-10-25|
+------+----+--------+----------+
only showing top 10 rows
```

Command took 0.53 seconds -- by prashantkumarpandey@gmail.com at 6/21/2020, 1:04:41 PM on MyTestCluster

Shift+Enter to run shortcuts

Cmd 8

```
airlinesDF.select("Origin", "Dest", "Distance", expr("to_date(concat(Year,Month, ...
```

Show cell

Cmd 9

```
airlinesDF.select("Origin", "Dest", "Distance", to_date(concat("Year","Month","DayofMonth"),'yyyyMMdd').alias("FlightDate")).show(10)
```

▸ (1) Spark Jobs

```
+------+----+--------+----------+
|Origin|Dest|Distance|FlightDate|
+------+----+--------+----------+
|   SAN| SFO|     447|1987-10-14|
|   SAN| SFO|     447|1987-10-15|
|   SAN| SFO|     447|1987-10-17|
|   SAN| SFO|     447|1987-10-18|
|   SAN| SFO|     447|1987-10-19|
|   SAN| SFO|     447|1987-10-21|
|   SAN| SFO|     447|1987-10-22|
|   SAN| SFO|     447|1987-10-23|
|   SAN| SFO|     447|1987-10-24|
|   SAN| SFO|     447|1987-10-25|
+------+----+--------+----------+
only showing top 10 rows
```

Command took 0.53 seconds -- by prashantkumarpandey@gmail.com at 6/21/2020, 1:07:01 PM on MyTestCluster

Shift+Enter to run shortcuts

ColumnsNotebook - Databricks

community.cloud.databricks.com/?o=5086207005927225#notebook/3635560979762597/command/3526529050390649

ColumnsNotebook (Python)

MyTestCluster | File | Edit | View: Code | Permissions | Run All | Clear | Publish | Comments | Runs | Revision history

Show cell

Cmd 8

```
airlinesDF.select("Origin", "Dest", "Distance", expr("to_date(concat(Year,Month, ...
```

Show cell

Cmd 9

```
airlinesDF.select("Origin", "Dest", "Distance", to_date(concat("Year","Month","DayofMonth"),'yyyyMMdd').alias("FlightDate")).show(10)
```

(1) Spark Jobs

```
+------+----+--------+----------+
|Origin|Dest|Distance|FlightDate|
+------+----+--------+----------+
|   SAN| SFO|     447|1987-10-14|
|   SAN| SFO|     447|1987-10-15|
|   SAN| SFO|     447|1987-10-17|
|   SAN| SFO|     447|1987-10-18|
|   SAN| SFO|     447|1987-10-19|
|   SAN| SFO|     447|1987-10-21|
|   SAN| SFO|     447|1987-10-22|
|   SAN| SFO|     447|1987-10-23|
|   SAN| SFO|     447|1987-10-24|
|   SAN| SFO|     447|1987-10-25|
+------+----+--------+----------+
only showing top 10 rows
```

Command took 0.53 seconds -- by prashantkumarpandey@gmail.com at 6/21/2020, 1:07:01 PM on MyTestCluster

1. DataFrame
2. Column
3. Built-in Functions

Shift+Enter to run shortcuts

```python

def parse_gender(gender):...


if __name__ == "__main__":
    spark = SparkSession \
        .builder \
        .appName("UDF Demo") \
        .master("local[2]") \
        .getOrCreate()

    logger = Log4j(spark)

    survey_df = spark.read \
        .option("header", "true") \
        .option("inferSchema", "true") \
        .csv("data/survey.csv")

    survey_df.show(10)

    parse_gender_udf = udf(parse_gender, StringType())
    survey_df2 = survey_df.withColumn("Gender", parse_gender_udf("Gender"))
    survey_df2.show(10)
```

```
31          .option("inferSchema", "true")
32          .csv("data/survey.csv")
33
34      survey_df.show(10)
35
36      parse_gender_udf = udf(parse_gender, StringType())
37      logger.info("Catalog Entry:")
38      [logger.info(f) for f in spark.catalog.listFunctions() if "parse_gender" in f.name]
39      survey_df2 = survey_df.withColumn("Gender", parse_gender_udf("Gender"))
40      survey_df2.show(10)
41
42      spark.udf.register("parse_gender_udf", parse_gender,StringType())
43      logger.info("Catalog Entry:")
44      [logger.info(f) for f in spark.catalog.listFunctions() if "parse_gender" in f.name]
45
46
```

```
31          .option("inferSchema", "true") \
32          .csv("data/survey.csv")
33
34      survey_df.show(10)
35
36      parse_gender_udf = udf(parse_gender, StringType())
37      logger.info("Catalog Entry:")
38      [logger.info(f) for f in spark.catalog.listFunctions() if "parse_gender" in f.name]
39      survey_df2 = survey_df.withColumn("Gender", parse_gender_udf("Gender"))
40      survey_df2.show(10)
41
42      spark.udf.register("parse_gender_udf", parse_gender,StringType())
43      logger.info("Catalog Entry:")
44      [logger.info(f) for f in spark.catalog.listFunctions() if "parse_gender" in f.name]
45      survey_df3 = survey_df.withColumn("Gender", expr("parse_gender_udf(Gender)"))
46
```

Miscellaneous

1. Quick method to create Dataframes
2. Adding monotonically increasing id
3. Using Case When Then transformation
4. Casting your columns
5. Adding columns to Dataframes
6. Dropping Columns
7. Dropping duplicate rows
8. Sorting Dataframes

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In [3]:

```python
import findspark
findspark.init()

import pyspark
from pyspark.sql import *
from pyspark.sql.functions import *
from pyspark.sql.types import *

spark = SparkSession.builder \
            .master("local[3]") \
            .appName("MiscDemo") \
            .getOrCreate()

data_list = [("Ravi", 28, 1, 2002),
             ("Abdul", 23, 5, 81),
             ("John", 12, 12, 6),
             ("Rosy", 7, 8, 63),
             ("Abdul", 23, 5, 81)
            ]
raw_df = spark.createDataFrame(data_list).toDF("name", "day", "month", "year")
raw_df.printSchema()
```

```
root
 |-- name: string (nullable = true)
 |-- day: long (nullable = true)
 |-- month: long (nullable = true)
 |-- year: long (nullable = true)
```

How to add monotonically increasing id?

Udemy

```python
import pyspark
from pyspark.sql import *
from pyspark.sql.functions import *
from pyspark.sql.types import *

spark = SparkSession.builder \
            .master("local[3]") \
            .appName("MiscDemo") \
            .getOrCreate()

data_list = [("Ravi", "28", "1", "2002"),
             ("Abdul", "23", "5", "81"),
             ("John", "12", "12", "6"),
             ("Rosy", "7", "8", "63"),
             ("Abdul", "23", "5", "81")
            ]
raw_df = spark.createDataFrame(data_list).toDF("name", "day", "month", "year").repartition(3)
raw_df.printSchema()
```

```
root
 |-- name: string (nullable = true)
 |-- day: string (nullable = true)
 |-- month: string (nullable = true)
 |-- year: string (nullable = true)
```

In [6]:

```python
df1 = raw_df.withColumn("id", monotonically_increasing_id())
df1.show()
```

```
+-----+---+-----+----+----------+
| name|day|month|year|        id|
+-----+---+-----+----+----------+
|Abdul| 23|    5|  81|         0|
| Ravi| 28|    1|2002| 8589934592|
|Abdul| 23|    5|  81| 8589934593|
| John| 12|   12|   6|17179869184|
| Rosy|  7|    8|  63|17179869185|
+-----+---+-----+----+----------+
```

In [8]:
```
df2 = df1.withColumn("year", expr("""
        case when year < 21 then year + 2000
        when year < 100 then year + 1900
        else year
        end"""))
df2.show()
```

1. Inline Cast
2. Change the Schema

```
+-----+---+-----+------+----------+
| name|day|month|  year|        id|
+-----+---+-----+------+----------+
|Abdul| 23|    5|1981.0|         0|
| Ravi| 28|    1|  2002| 8589934592|
|Abdul| 23|    5|1981.0| 8589934593|
| John| 12|   12|2006.0|17179869184|
| Rosy|  7|    8|1963.0|17179869185|
+-----+---+-----+------+----------+
```

```
| John| 12|   12|2006.0|17179869184|
| Rosy|  7|    8|1963.0|17179869185|
+-----+---+-----+------+-----------+
```

In [9]:

```
df3 = df1.withColumn("year", expr("""
        case when year < 21 then cast(year as int) + 2000
        when year < 100 then cast(year as int) + 1900
        else year
        end""" ))
df3.show()
```

```
+-----+---+-----+----+-----------+
| name|day|month|year|         id|
+-----+---+-----+----+-----------+
|Abdul| 23|    5|1981|          0|
| Ravi| 28|    1|2002| 8589934592|
|Abdul| 23|    5|1981| 8589934593|
| John| 12|   12|2006|17179869184|
| Rosy|  7|    8|1963|17179869185|
+-----+---+-----+----+-----------+
```

1. Inline Cast
2. Change the Schema

In [ ]:

```
df4 = df1.withColumn("year", expr("""
        case when year < 21 then year + 2000
        when year < 100 then cast(year + 1900
        else year
        end""" ))
df4.show()
```

```
root
 |-- name: string (nullable = true)
 |-- day: string (nullable = true)
 |-- month: string (nullable = true)
 |-- year: integer (nullable = true)
 |-- id: long (nullable = false)
```

In [14]:

```
df5 = df1.withColumn("day", col("day").cast(IntegerType())) \
         .withColumn("month", col("month").cast(IntegerType())) \
         .withColumn("year", col("year").cast(IntegerType()))

df6 = df5.withColumn("year", expr("""
          case when year < 21 then year + 2000
          when year < 100 then year + 1900
          else year
          end"""))
df6.show()
```

```
+-----+---+-----+----+-----------+
| name|day|month|year|         id|
+-----+---+-----+----+-----------+
|Abdul| 23|    5|1981|          0|
| Ravi| 28|    1|2002| 8589934592|
|Abdul| 23|    5|1981| 8589934593|
| John| 12|   12|2006|17179869184|
| Rosy|  7|    8|1963|17179869185|
+-----+---+-----+----+-----------+
```

1. Inline Cast
2. Change the Schema

```
+-----+---+-----+----+----------+
|Abdul| 23|    5|1981|         0|
| Ravi| 28|    1|2002| 8589934592|
|Abdul| 23|    5|1981| 8589934593|
| John| 12|   12|2006|17179869184|
| Rosy|  7|    8|1963|17179869185|
+-----+---+-----+----+----------+
```

In [15]:

```
df7 = df5.withColumn("year", \
                     when(col("year") < 21, col("year") + 2000) \
                     .when(col("year") < 100, col("year") + 1900) \
                     .otherwise(col("year")))
df7.show()
```

```
+-----+---+-----+----+----------+
| name|day|month|year|        id|
+-----+---+-----+----+----------+
|Abdul| 23|    5|1981|         0|
| Ravi| 28|    1|2002| 8589934592|
|Abdul| 23|    5|1981| 8589934593|
| John| 12|   12|2006|17179869184|
| Rosy|  7|    8|1963|17179869185|
+-----+---+-----+----+----------+
```

Run Code

In [16]:
```
df8 = df7.withColumn("dob", expr("to_date(concat(day,'/',month,'/',year), 'd/M/y')"))
df8.show()
```

```
+-----+---+-----+----+-----------+----------+
| name|day|month|year|         id|       dob|
+-----+---+-----+----+-----------+----------+
|Abdul| 23|    5|1981|          0|1981-05-23|
| Ravi| 28|    1|2002| 8589934592|2002-01-28|
|Abdul| 23|    5|1981| 8589934593|1981-05-23|
| John| 12|   12|2006|17179869184|2006-12-12|
| Rosy|  7|    8|1963|17179869185|1963-08-07|
+-----+---+-----+----+-----------+----------+
```

In [17]:
```
df9 = df7.withColumn("dob", to_date(expr("concat(day,'/',month,'/',year)"), 'd/M/y'))
df9.show()
```

```
+-----+---+-----+----+-----------+----------+
| name|day|month|year|         id|       dob|
+-----+---+-----+----+-----------+----------+
|Abdul| 23|    5|1981|          0|1981-05-23|
| Ravi| 28|    1|2002| 8589934592|2002-01-28|
|Abdul| 23|    5|1981| 8589934593|1981-05-23|
| John| 12|   12|2006|17179869184|2006-12-12|
| Rosy|  7|    8|1963|17179869185|1963-08-07|
+-----+---+-----+----+-----------+----------+
```

In [16]:
```
df8 = df7.withColumn("dob", expr("to_date(concat(day,'/',month,'/',year), 'd/M/y')"))
df8.show()
```

```
+-----+---+-----+----+-----------+----------+
| name|day|month|year|         id|       dob|
+-----+---+-----+----+-----------+----------+
|Abdul| 23|    5|1981|          0|1981-05-23|
| Ravi| 28|    1|2002| 8589934592|2002-01-28|
|Abdul| 23|    5|1981| 8589934593|1981-05-23|
| John| 12|   12|2006|17179869184|2006-12-12|
| Rosy|  7|    8|1963|17179869185|1963-08-07|
+-----+---+-----+----+-----------+----------+
```

In [20]:
```
df9 = df7.withColumn("dob", to_date(expr("concat(day,'/',month,'/',year)"), 'd/M/y')) \
        .drop("day", "month", "year") \
        .dropDuplicates(["name", "dob"]) \
        .sort(expr("dob desc"))
df9.show()
```

```
+-----+-----------+----------+
| name|         id|       dob|
+-----+-----------+----------+
| Rosy|17179869185|1963-08-07|
|Abdul|          0|1981-05-23|
| Ravi| 8589934592|2002-01-28|
| John|17179869184|2006-12-12|
+-----+-----------+----------+
```